

Math Hard Question 3

1

★ Mark for Review

ABC

A research manager selected 2 random samples of ovens of a certain type to estimate the average amount of time this type of oven takes to preheat to $325^{\circ}F$. The research manager recorded the amount of time, in minutes, each oven takes to preheat to $325^{\circ}F$.

Based on the first sample, the manager estimated an average of 14.2 minutes, with a margin of error of 1 minute. Based on the second sample, the estimate was 14.4 minutes, with a margin of error of 2.3 minutes.

Assuming the margins of error were calculated the same way, which of the following best explains why the first sample had a smaller margin of error?

A. The first sample contained fewer ovens than the second sample.

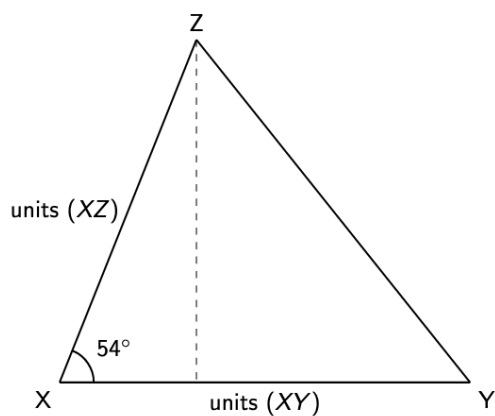
B. The first sample contained more ovens than the second sample.

C. The first sample took less time on average to preheat.

D. The first sample took more time on average to preheat.

2

★ Mark for Review

ABC

In triangle XYZ , $\angle X = 54^\circ$, $XY = 26$ units, $XZ = 19$ units. What is the area, in square units, of triangle XYZ ?

A. 247

B. 494

C. $247\sin 54^\circ$ D. $494\sin 54^\circ$ **3**

★ Mark for Review

ABC

In the figure, $\triangle ABC$ is inscribed in a semicircle with diameter $AC = 10$. Point B lies on the semicircle. If $AB = 6$ and $BC = x$, find x . Give your answer in simplest radical form.

A. $\sqrt{36}$

B. 8

C. $\sqrt{44}$

D. $\sqrt{64}$

4

★ Mark for Review

ABC

A parabola is given by

$$y = x^2 - 4x + 5,$$

and a circle by

$$x^2 + y^2 = 25.$$

The parabola and circle intersect at points P and Q . Find the distance between P and Q .

5

★ Mark for Review

ABC

Solve the equation:

$$\sqrt{6r} = \sqrt{r^2 - 16}$$

What is the product of all solutions to the above equation?

A. 4

B. 6

C. 8

D. 12